Q1. A hospital dashboard displays patient age using point size and blood pressure using point color. Elderly patients appear larger, while blood pressure is represented using a blue-to-red gradient. Which visualization principle is primarily violated?

A. Position should not encode quantitative values.
B. Point size is unsuitable for ordered quantitative comparisons.
C. Color gradients cannot represent quantitative values.
D. Age should always be mapped to the x-axis.

Q2. A financial analyst uses different shapes to represent monthly stock prices over time. Users struggle to identify trends. Which mapping would improve quantitative interpretation?

A. Map stock prices to shape.
B. Map stock prices to color hue.
C. Map stock prices to vertical position.
D. Map stock prices to transparency.

Q3. A researcher maps customer satisfaction levels (Very Poor, Poor, Fair, Good, Excellent) to random colors without ordering. The visualization becomes difficult to interpret because

A. Satisfaction is nominal.
B. Satisfaction is ordinal and requires ordered aesthetics.
C. Random colors increase contrast.
D. Color hue automatically implies ordering.

Q4. A climatologist maps rainfall intensity simultaneously to point size and color intensity. The visualization becomes misleading because

A. Redundant encoding may exaggerate perceived differences.
B. Color cannot represent quantitative variables.
C. Point size cannot encode quantitative data.
D. Multiple aesthetics always improve interpretation.

Q5. An analyst represents employee departments using a continuous color gradient. Which issue arises?

A. Continuous scales imply ordering where none exists.
B. Gradients improve categorical distinction.
C. Departments are quantitative.
D. Cartesian coordinates become invalid.

Q6. A scatter plot maps income to x-position and education level to y-position. Gender is represented using color. Which aesthetic represents nominal data?

A. x-position
B. y-position
C. Color
D. Position

Q7. A visualization maps annual sales to point transparency instead of position. Why is this a poor choice?

A. Transparency has low perceptual accuracy.
B. Transparency cannot encode variables.
C. Transparency represents categories only.
D. Position should only represent nominal data.

Q8. A chart uses texture patterns instead of colors to distinguish land-use categories. The primary advantage is

A. Improved quantitative precision
<u>B. Better accessibility for color-blind users</u>
C. Reduced memory usage
D. Faster rendering speed

Q9. A dataset contains Country, GDP, Population, and Continent. Which variable should ideally be encoded using shape?

A. GDP
B. Population
<u>C. Continent</u>
D. GDP Growth

Q10. Which visualization mapping produces the highest perceptual accuracy for comparing numerical values?

A. Area
B. Color saturation
<u>C. Position along a common scale</u>
D. Transparency

Q11. A data scientist uses a logarithmic scale for household income. Which reason best justifies this choice?

A. Income contains many categories.
<u>B. Income spans several orders of magnitude.</u>
C. Log scales improve categorical comparison.
D. Logarithmic scales remove outliers.

Q12. A temperature visualization assigns identical colors to values between 20°C and 25°C. The scale primarily reduces

<u>A. Precision</u>
B. Coordinate transformation
C. Scatter density
D. Position accuracy

Q13. A continuous variable is mapped to five discrete colors. Which problem is introduced?

A. False precision
<u>B. Information loss</u>
C. Axis inversion
D. Overplotting

Q14. A scatter plot uses identical symbol sizes for all observations while mapping sales to color intensity. Which aesthetic is unused?

A. Position
<u>B. Size</u>
C. Color
D. Shape

Q15. When transforming data values into visual positions, the mapping function is called

A. Coordinate system
B. Scale
C. Axis
D. Legend

Q16. A scientist rotates Cartesian coordinates by 45°. Which property remains unchanged?

A. Relative distances
B. Scale labels
C. Color palette
D. Shape encoding

Q17. A visualization requires comparison of radial distances from a central point. Which coordinate system is appropriate?

A. Cartesian
B. Polar
C. Parallel
D. Ternary

Q18. A business analyst compares quarterly revenues using pie slices arranged radially. Which issue is most likely?

A. Difficulty comparing angles accurately
B. Cartesian distortion
C. Incorrect logarithmic transformation
D. Missing legends

Q19. A map uses latitude and longitude. These coordinates are examples of

A. Polar
B. Cartesian
C. Geographic coordinate system
D. Logarithmic coordinates

Q20. Why are Cartesian coordinates preferred for scatter plots?

A. They improve color perception.
B. They preserve independent quantitative dimensions.
C. They require fewer data points.
D. They eliminate outliers.

Q21. A microbiologist visualizes bacterial counts ranging from 100 to 10 million. Which axis is most suitable?

A. Linear
B. Polar
C. Logarithmic
D. Circular

Q22. Which statement about logarithmic axes is TRUE?

A. Equal distances represent equal differences.
B. Equal distances represent equal ratios.
C. Values cannot be negative.
D. Both B and C.

Q23. A student uses a log axis for exam scores between 60 and 100. The visualization becomes

A. More informative
<u>B. Less meaningful because data span is small</u>
C. Easier to compare
D. Better for categorical data

Q24. Applying nonlinear scaling mainly helps

A. Increase categorical distinction
<u>B. Compress skewed numerical distributions</u>
C. Improve color accessibility
D. Eliminate missing values

Q25. Which transformation best linearizes exponential growth?

A. Square root
<u>B. Logarithm</u>
C. Reciprocal
D. Mean normalization

Q26. Which visualization naturally uses polar coordinates?

A. Heatmap
<u>B. Radar chart</u>
C. Scatter plot
D. Histogram

Q27. A wind-direction visualization requires angle and distance measurements. Which coordinate system is ideal?

A. Cartesian
<u>B. Polar</u>
C. Geographic
D. Matrix

Q28. Pie charts primarily encode data through

A. Length
B. Area
<u>C. Angle</u>
D. Transparency

Q29. Why are bar charts generally preferred over pie charts?

<u>A. Bars compare lengths more accurately than angles.</u>
B. Pie charts cannot display percentages.
C. Bars require fewer colors.
D. Pie charts cannot represent categorical data.

Q30. A circular timeline is chosen for seasonal events because

<u>A. Seasons are cyclic.</u>
B. Cartesian axes are unavailable.
C. Polar coordinates increase precision.
D. Colors require circular layouts.

Q31. A disease severity map uses green→yellow→red colors. This represents

A. Qualitative scale
<u>B. Sequential scale</u>
C. Diverging scale
D. Nominal scale

Q32. A map shows temperature anomalies centered at zero. Which palette is most suitable?

A. Sequential
<u>B. Diverging</u>
C. Qualitative
D. Random

Q33. Which color scale best distinguishes political parties?

A. Sequential
<u>B. Qualitative</u>
C. Diverging
D. Continuous grayscale

Q34. A visualization assigns identical colors to two unrelated categories. This primarily affects

A. Accuracy
<u>B. Distinguishability</u>
C. Axis scaling
D. Coordinate mapping

Q35. Which palette is inappropriate for categorical variables?

A. Qualitative
<u>B. Sequential</u>
C. Distinct hues
D. Random but unique colors

Q36. A transportation map uses different colors for bus, metro, train, and tram routes. Color serves to

A. Encode quantitative values
<u>B. Distinguish categories</u>
C. Represent uncertainty
D. Encode rankings

Q37. Why should adjacent categories use highly distinguishable colors?

<u>A. Improve perceptual separation</u>
B. Reduce axis labels
C. Compress data
D. Improve logarithmic scaling

Q38. A visualization uses red and green only. Which users may face difficulty?

<u>A. Color-blind viewers</u>
B. Statisticians
C. Designers
D. Programmers

Q39. A qualitative palette should primarily maximize

A. Saturation ordering

B. Hue differences
C. Brightness ordering
D. Transparency

Q40. When many categories exist, the best strategy is
A. Use random colors.
B. Combine color with shapes.
C. Use grayscale only.
D. Increase transparency.

Q41. A heatmap uses darker shades for higher values. This is an example of
A. Sequential color encoding
B. Qualitative mapping
C. Shape encoding
D. Nominal mapping

Q42. Which palette best emphasizes values above and below average?
A. Qualitative
B. Sequential
C. Diverging
D. Rainbow

Q43. Using rainbow palettes for ordered data may confuse viewers because
A. Hue changes lack perceptual ordering.
B. Rainbow colors are inaccessible.
C. They reduce rendering speed.
D. They eliminate legends.

Q44. A financial dashboard colors profits blue and losses orange around zero. This is
A. Sequential mapping
B. Diverging mapping
C. Qualitative mapping
D. Nominal mapping

Q45. Which aesthetic best complements color for quantitative heatmaps?
A. Shape
B. Position
C. Size
D. Transparency

Q46. A dashboard displays all bars gray except the current month in red. The red color is used to
A. Encode categories
B. Highlight important information
C. Represent magnitude
D. Encode uncertainty

Q47. Overusing highlight colors causes
A. Improved focus
B. Reduced emphasis
C. Better quantitative comparison

D. Improved accessibility

Q48. A researcher wants readers to notice only failed experiments. Which color strategy is best?
A. Bright colors for all experiments
B. Neutral colors with failures highlighted
C. Sequential palette
D. Random colors

Q49. A visualization will be printed in grayscale. Which palette choice is most appropriate?
A. Random hues
B. Equal luminance colors
C. High luminance contrast
D. Rainbow palette

Q50. A global health dashboard must be interpretable by users with various forms of color vision deficiency. Which design decision is best?
A. Use only red and green.
B. Use color-blind-friendly palettes and redundant encodings.
C. Use maximum saturation for every category.
D. Avoid legends.